

Vertex Reconstruction using (GraphNeT-based) GNN for ANNIE

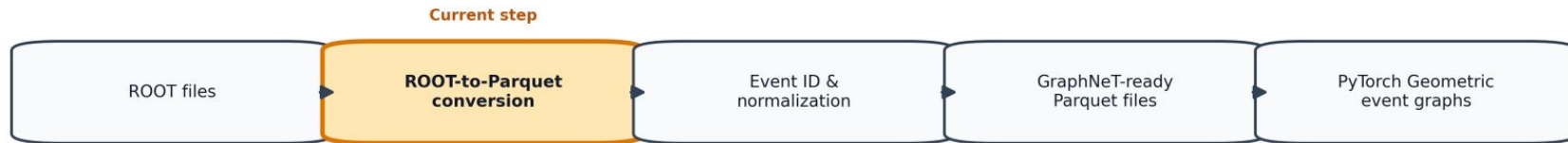
Efthymios Grizanis, Chrysovalantis Karagiannis

NCSR Demokritos

Vertex Reconstruction

- The interaction vertex corresponds to the 3D position where the neutrino interaction occurs inside the detector . In this work , the target vertex position is given by the Monte Carlo truth coordinates stored in the ROOT files : `trueVtxX` , `trueVtxY` , `trueVtxZ` . Vertex reconstruction is important for event reconstruction , since the vertex defines the event origin and provides essential geometrical information about the interaction .
- We train a Graph Neural Network to predict the vertex coordinates : `position_x_pred` , `position_y_pred` , `position_z_pred` . Supervised learning is used based on the true vertex coordinates.
- The DynEdge architecture from GraphNeT framework was used (the same algorithm used for energy reconstruction)

Preprocessing : ROOT to Parquet Conversion



The root files (output of PhaseITreeMaker) contain the event information used for the vertex reconstruction . Each event contains a variable number of detector hits/pulses and during the conversion we select the branches that we need :

1. Hit-level detector variables :
digitX , digitY , digitZ , digitT , digitQ

2. Monte-Carlo / Truth vertex coordinates :
trueVtxX , trueVtxY , trueVtxZ

The output of this step is a set of Parquet files that will be used in the next preprocessing steps .

After the ROOT-to-Parquet conversion , each event is assigned a unique eventNumber . The eventNumber acts as the unique ID of the event across all the Parquet Files . This ID is used later to match all information belonging to the same event .

Preprocessing: Event IDs , Normalization & Weights

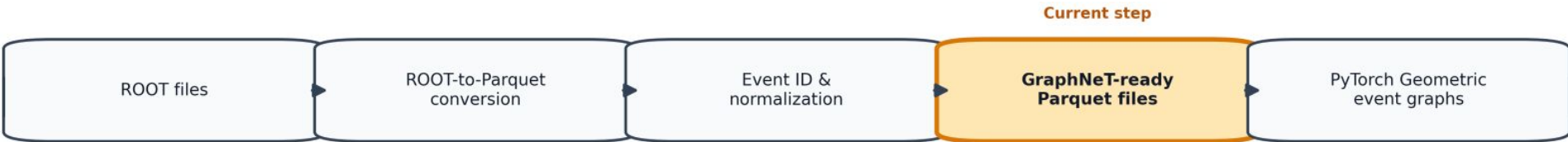


Next , the mean and the standard deviation of the input features are computed and we use them to normalize the five input features : digitX,digitY,digitZ,digitT,digitQ. Normalization brings the input features to a similar scale before training. This prevents features with larger numerical values from dominating the training and helps the gradients become more stable during backpropagation.

$$x_{\text{norm}} = (x - \mu) / \sigma$$

μ and σ are computed separately for each input feature using all hit values from the preprocessed Parquet files. The resulting values are stored and reused during training, validation, testing and prediction.

Preprocessing: GraphNeT Parquet Dataset



In this step , we reorganize the Parquet files into the structure expected by GraphNet. The output directory contains three aligned folders :

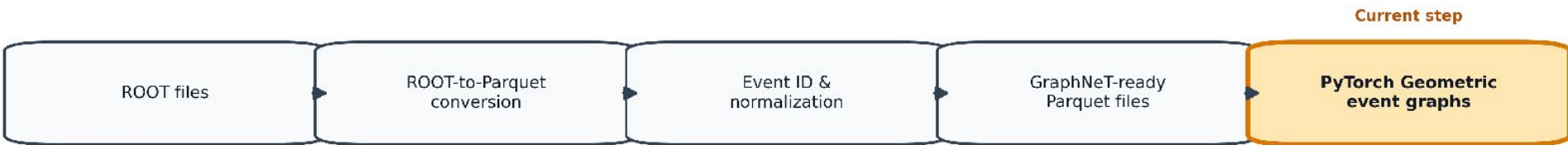
- ❖ features -> hit-level input variables
- ❖ truth -> target values
- ❖ weights -> per-event weights used in the loss

For each file index i , the script creates :

features_ i .parquet
truth_ i .parquet
weights_ i .parquet

The same eventNumber is preserved across the three files, allowing the information of each event to be matched correctly.

Graph Creation



Each event is converted into an independent graph.

Detector hits/pulses become graph nodes with features:
digitX,digitY,digitZ,digitT,digitQ

Edges are created with K-nearest neighbors using $k=5$, connecting nearby hits for message passing.

The resulting graphs are saved as PyTorch Geometric .pt files and used as input to DynEdge

Graph Neural Network (GNN) from GraphNeT

Input features for each detector hit :

digitX,digitY,digitZ

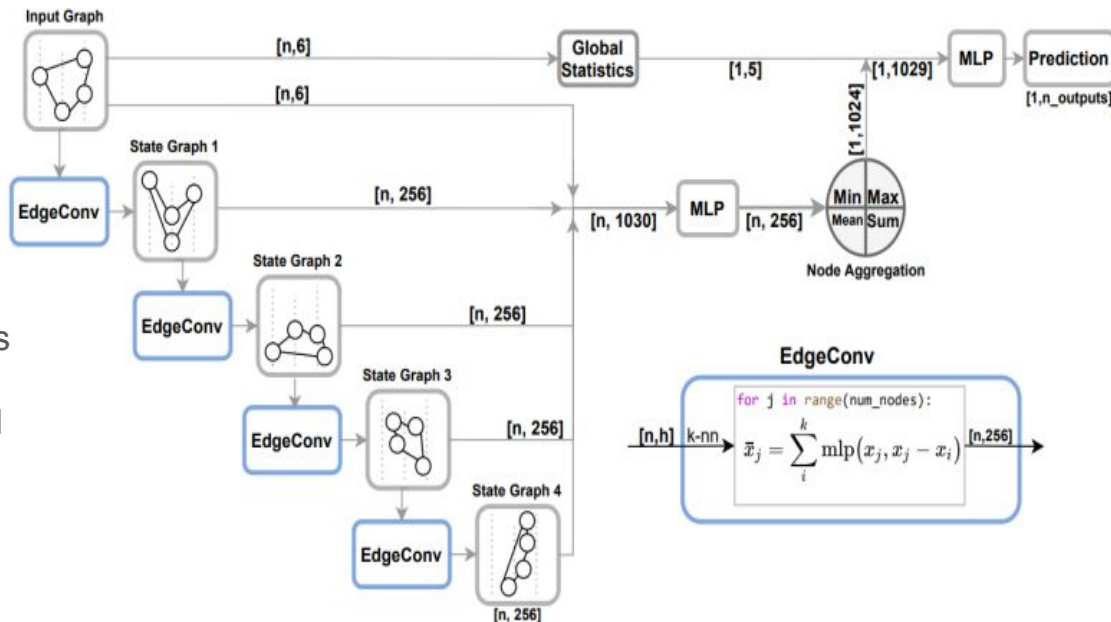
digitT

digitQ

Model setup :

- Architecture : DynEdge
- Graph construction : K-Nearest Neighbors
- Number of neighbors : $k = 5$
- Pooling operations: sum pooling removed
- Loss function: Logcosh loss
- Optimizer: AdamW
- Learning rate: Piecewise linear learning rate : $5 \times 10^{-6} \rightarrow 1 \times 10^{-4} \rightarrow 5 \times 10^{-6}$
- Batch size: 32 , 16
- Max epochs: 20 , 30
- Early stopping patience : 5

Output: one reconstructed vertex position (X, Y, Z) per event graph



DynEdge architecture

Files used & Event Selection

- Files used:

/pnfs/annie/persistent/simulations/wcsim/G1810a0211a/standard/tank/

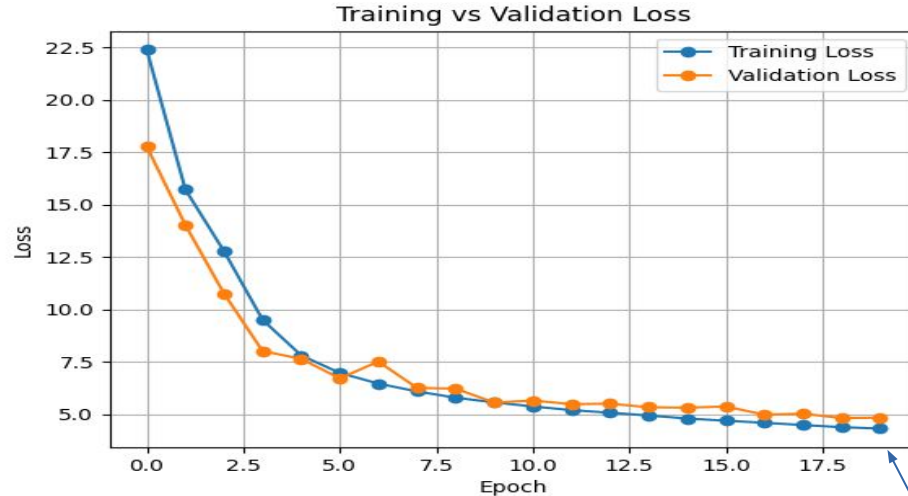
- Cuts applied:

EventSelector configuration:

verbosity 0
MCPMTVolCut 0
MCFVCut 1
MCMRDCut 0
MCPiKCt 0
MCIsMuonCut 1
MRDRecoCut 0
RecoPMTVolCut 0
RecoFVCut 0
NoVeto 1
NHitCut 1
PromptTrigOnly 1
SaveStatusToStore 1
Triggerword -1

Training and Vertex Prediction Results

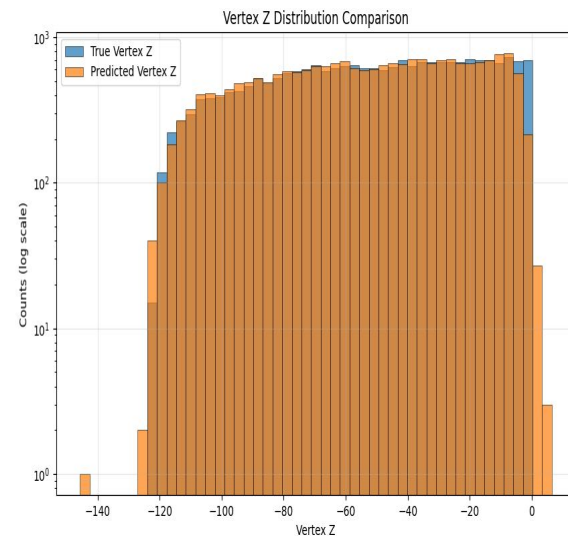
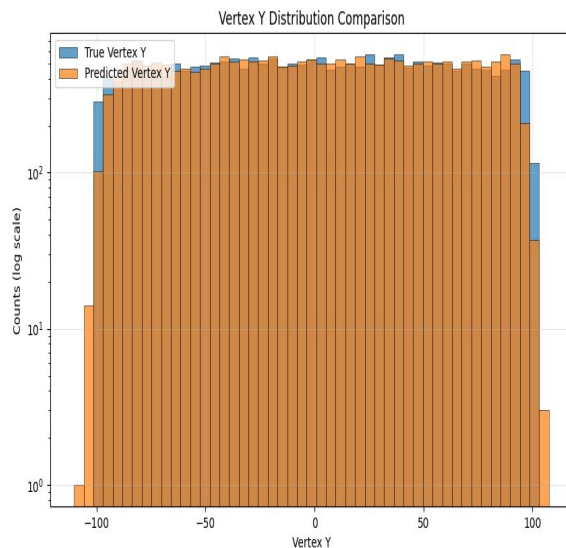
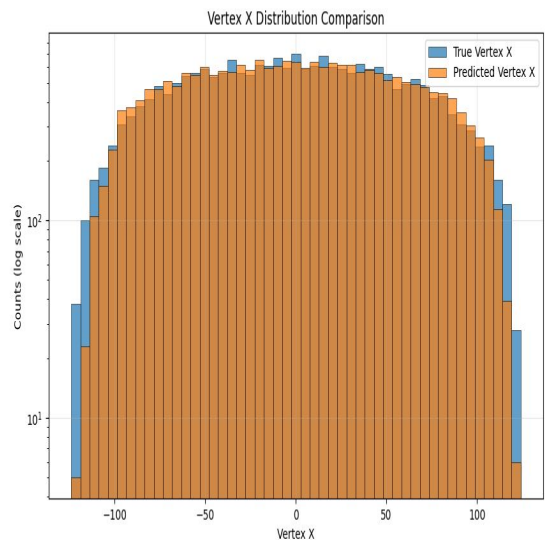
- Model was trained using ~218 114 events with a ~80%-10%-10% split for training ,validation and testing.
- The results shown refer to the prediction on the independent test set of 21,812 events.
- We are gonna start off with batch size = 16 & 20 epochs .Early stopping is applied to avoid continuing training if the validation loss no longer decreases. We also trained the model with larger batch sizes , but the results were slightly worse.



The model used for this prediction is the final saved model after training because no explicit best-epoch checkpoint was used in this run.

1D Distributions

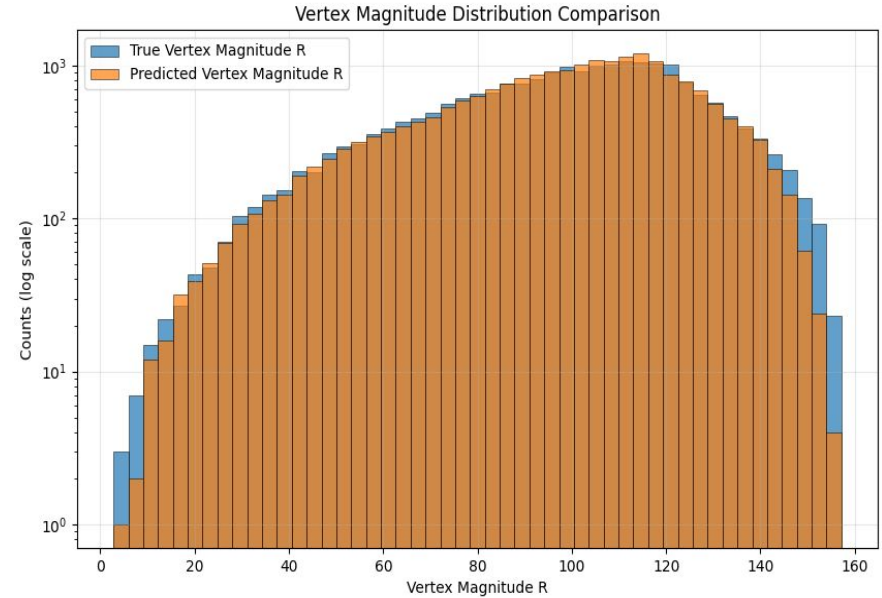
- Plotting the 1D distributions of the different vertex position components and comparing with the predicted vertex position by the GNN.



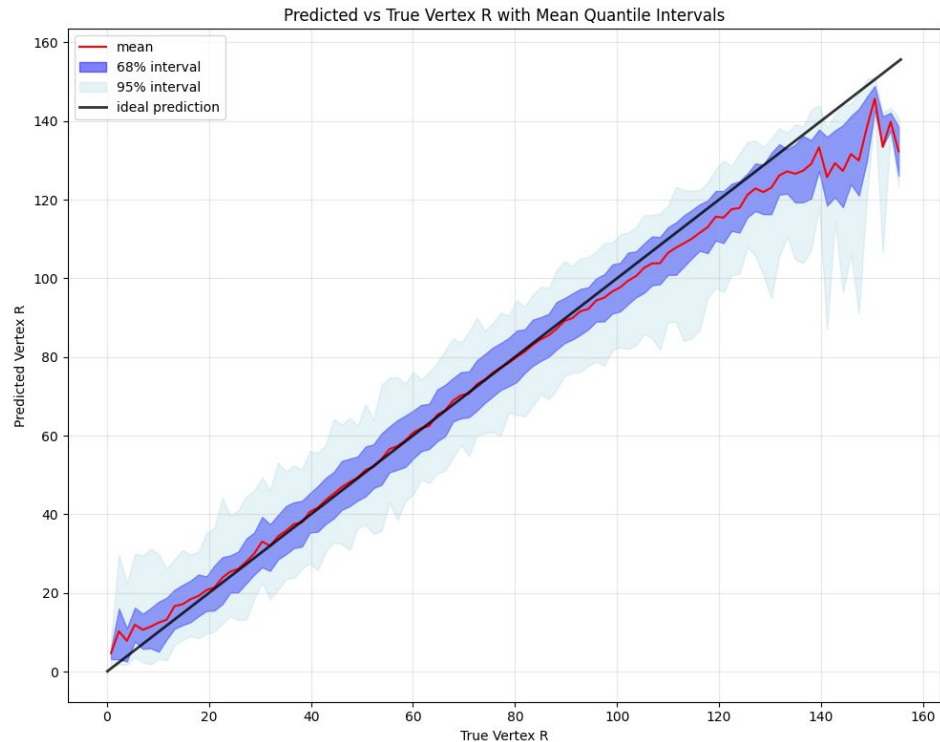
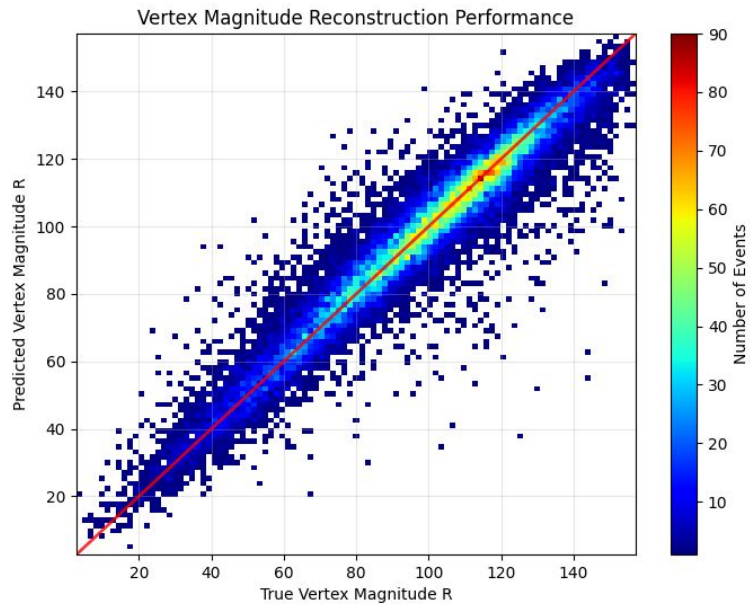
1D Distributions

The radial vertex coordinate is defined as :

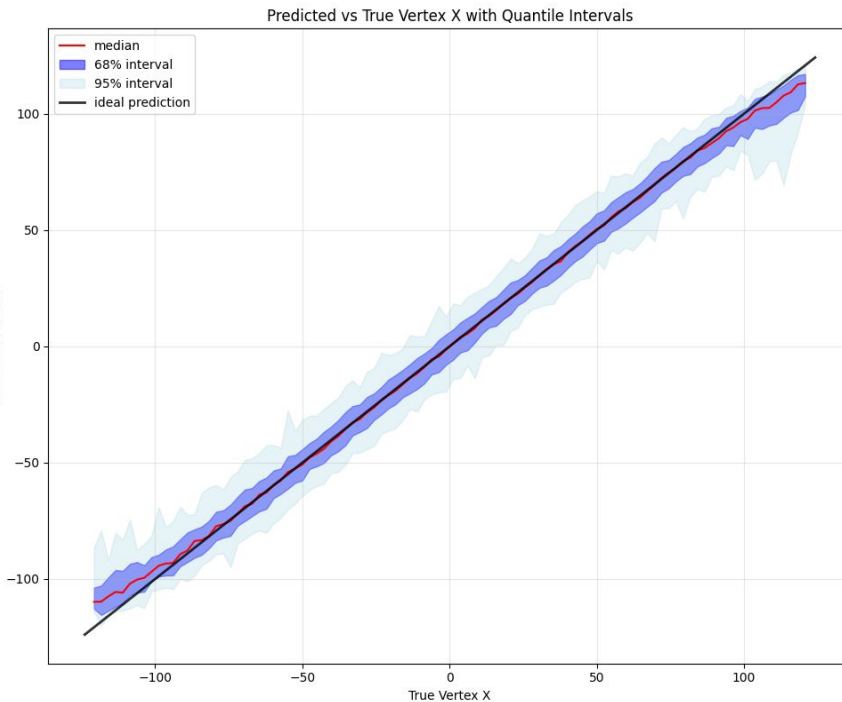
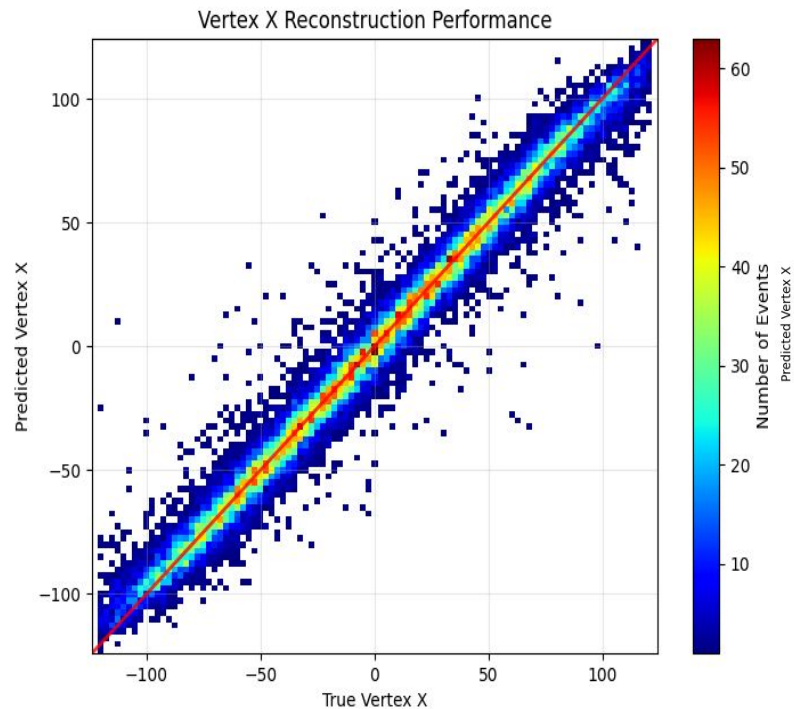
$$R = \sqrt{X^2 + Y^2 + Z^2}$$



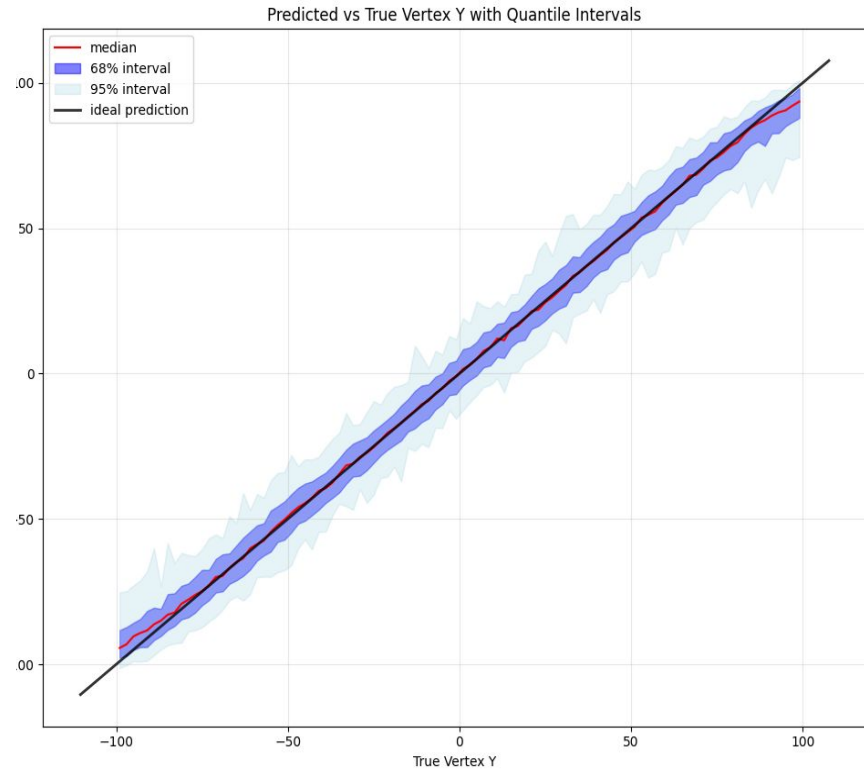
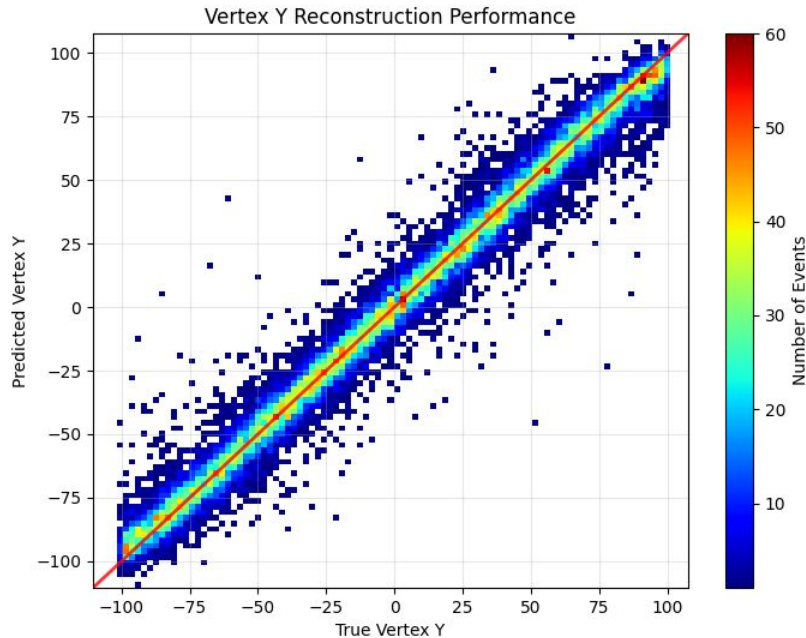
Vertex R Reconstruction and Mean Plots



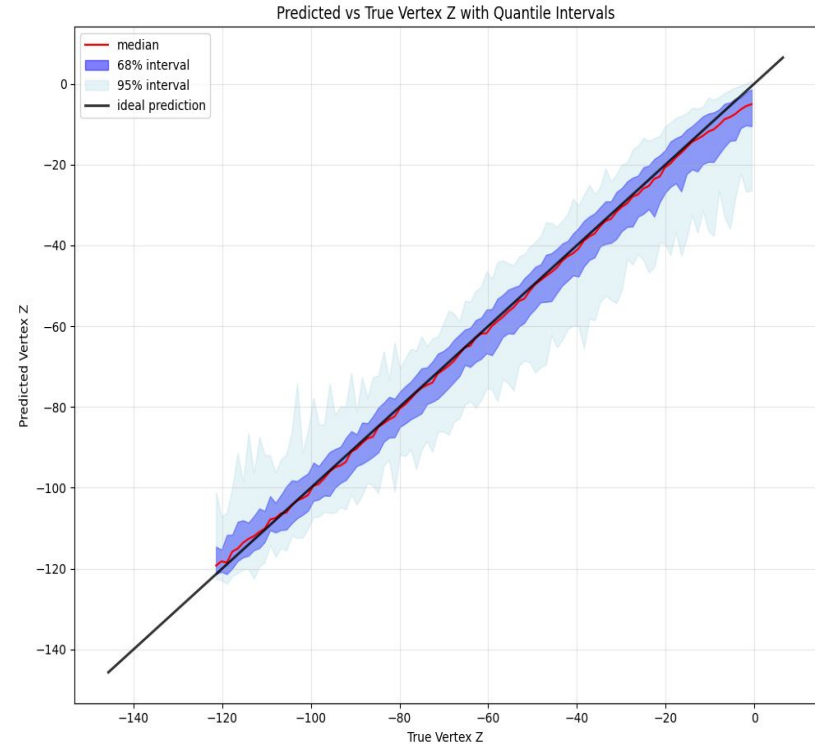
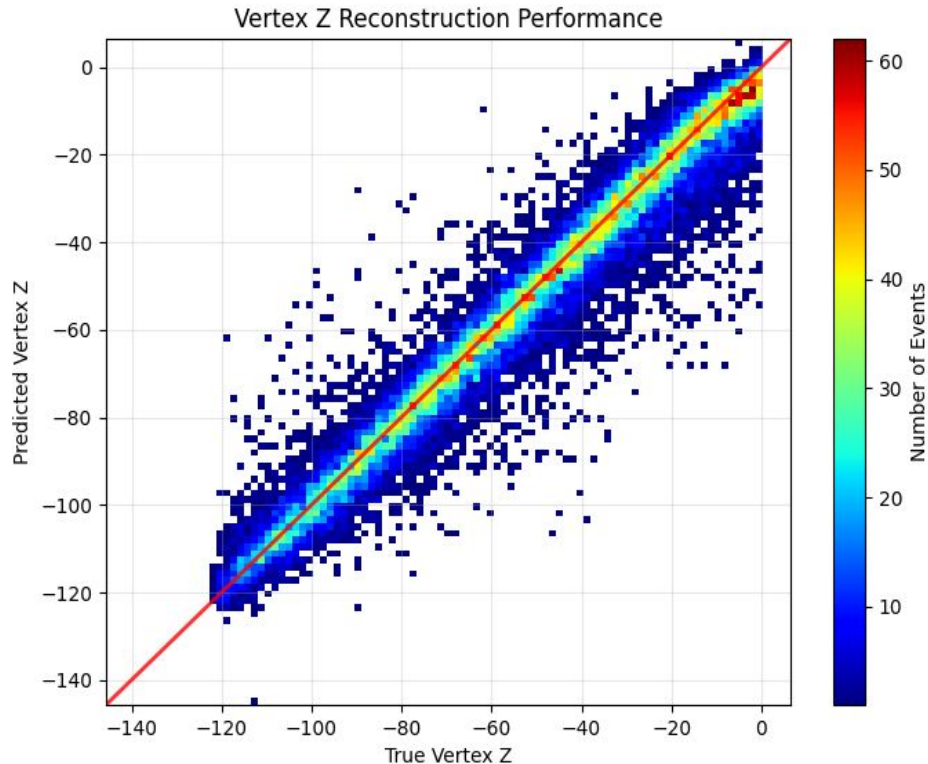
Vertex X Reconstruction and Mean plots



Vertex Y Reconstruction and Mean plots

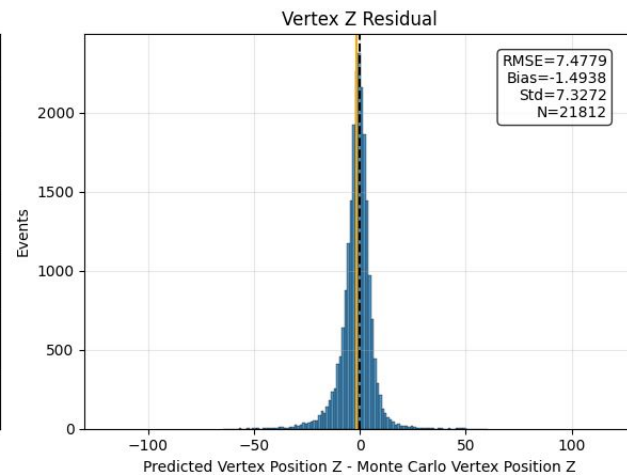
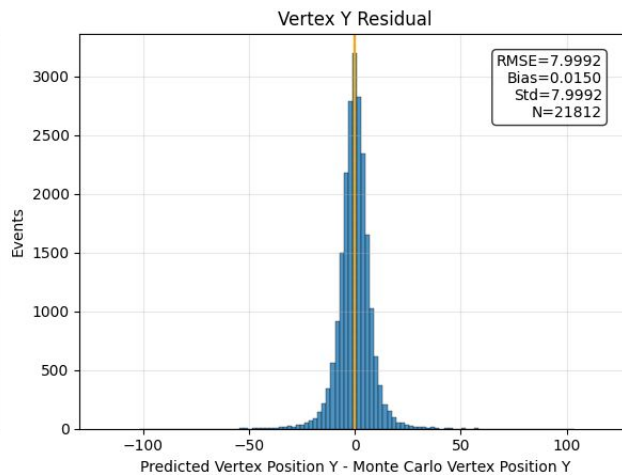
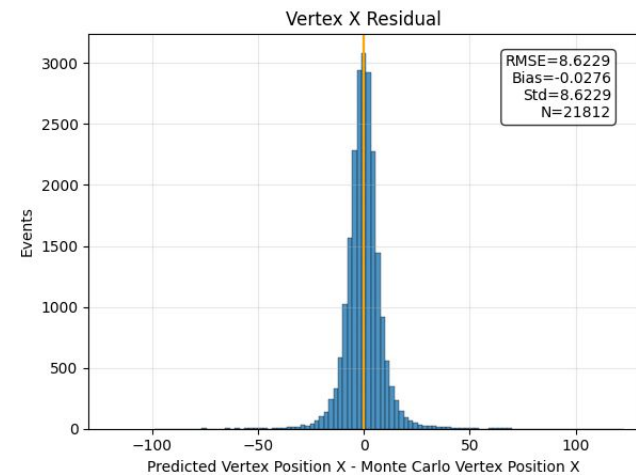


Vertex Z Reconstruction and Mean plots

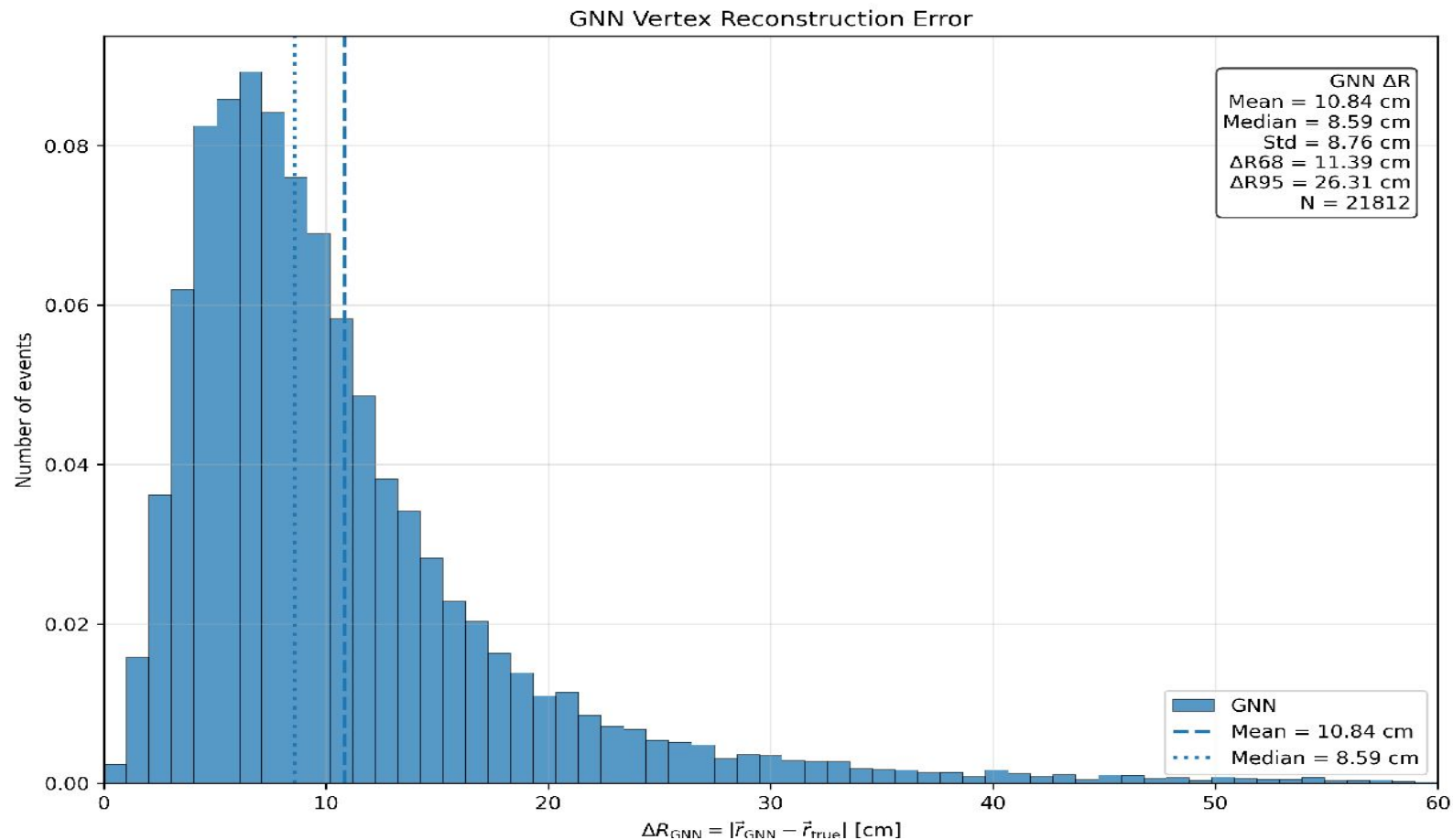


Residual

Vertex Position Reconstruction Residuals



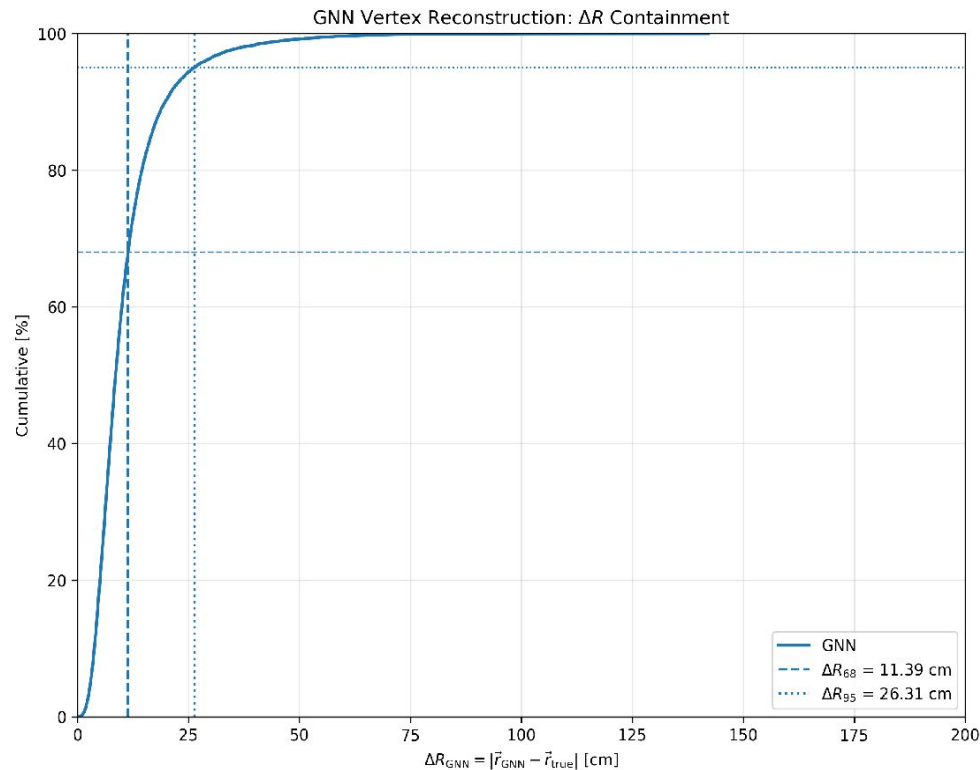
ΔR GNN



ΔR Containment: GNN

The containment curve shows the percentage of events with reconstruction error below a given ΔR value.

ΔR_{68} and ΔR_{95} are the error thresholds that contain 68% and 95% of the selected events.



Training and Vertex Prediction Results

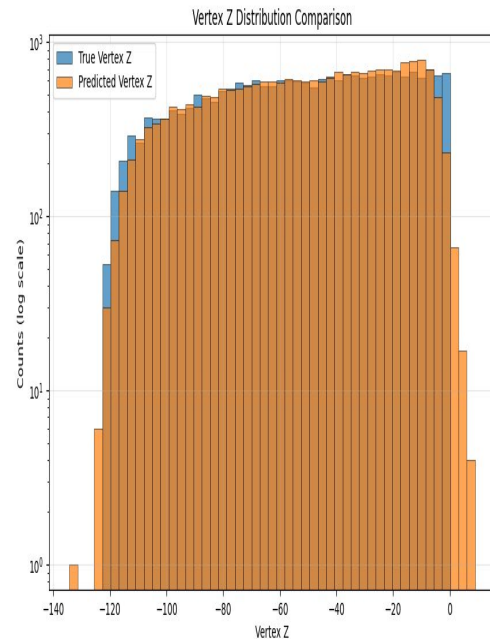
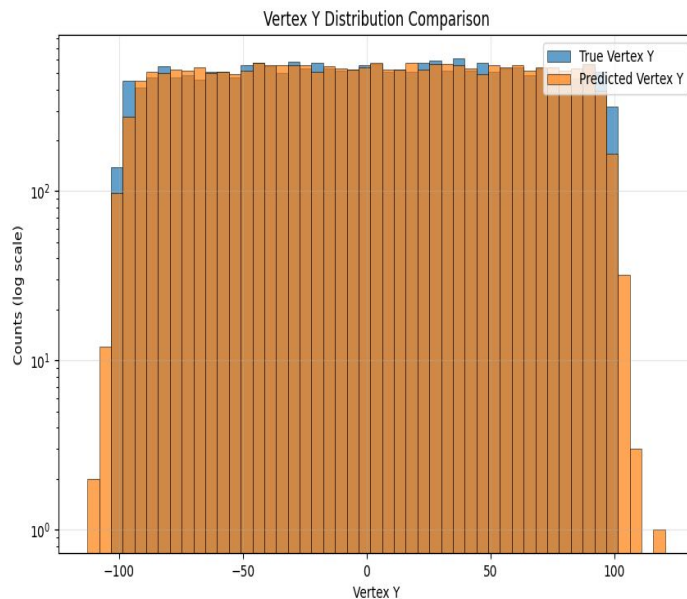
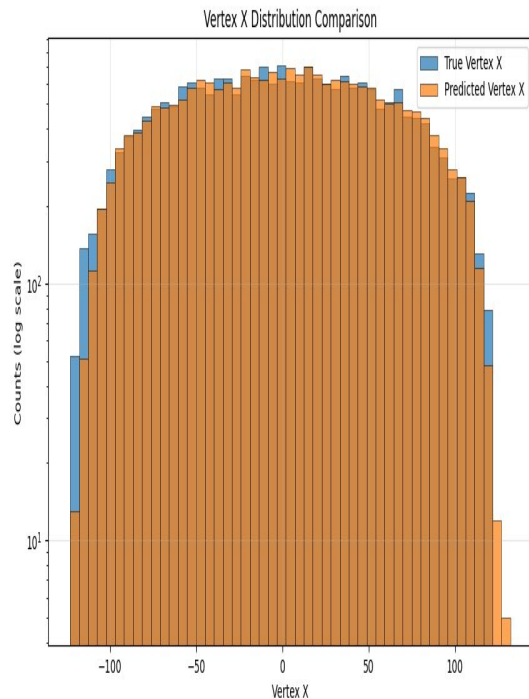
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Distribution

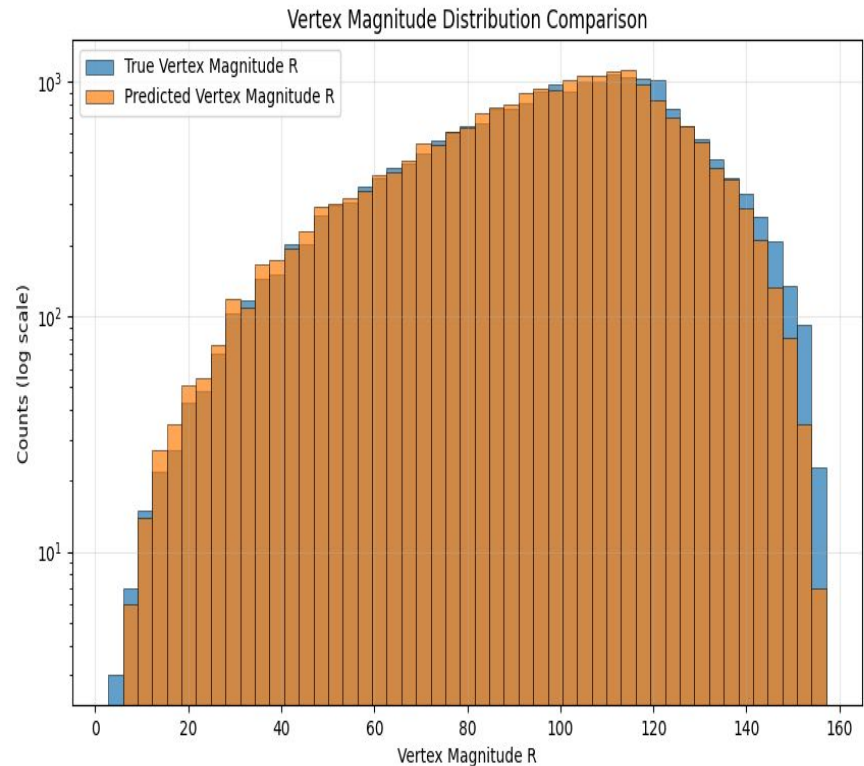
Vertex Coordinate Distributions : GNN Prediction vs Monte Carlo Truth



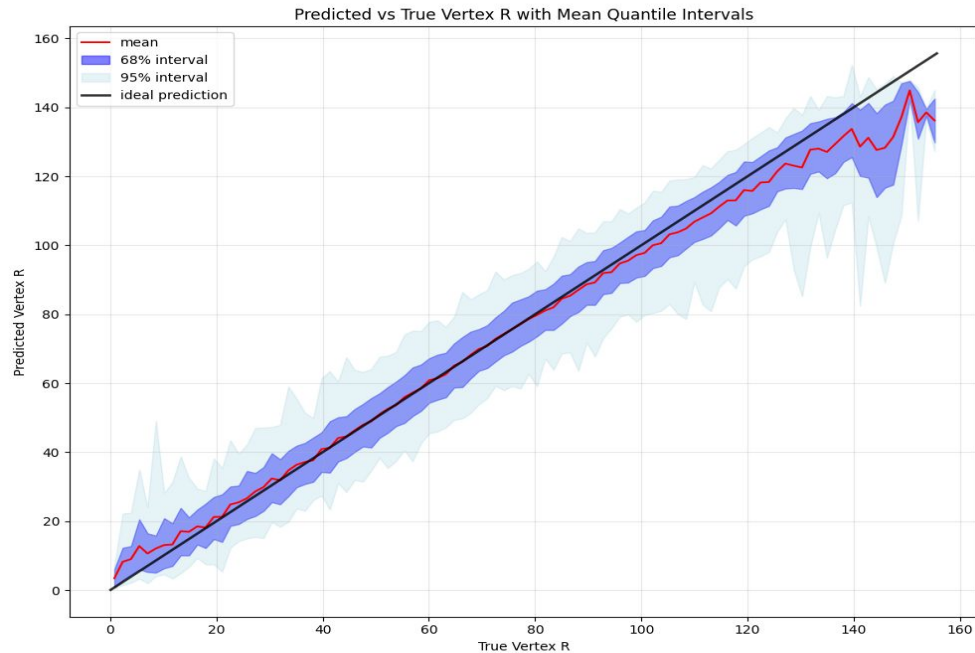
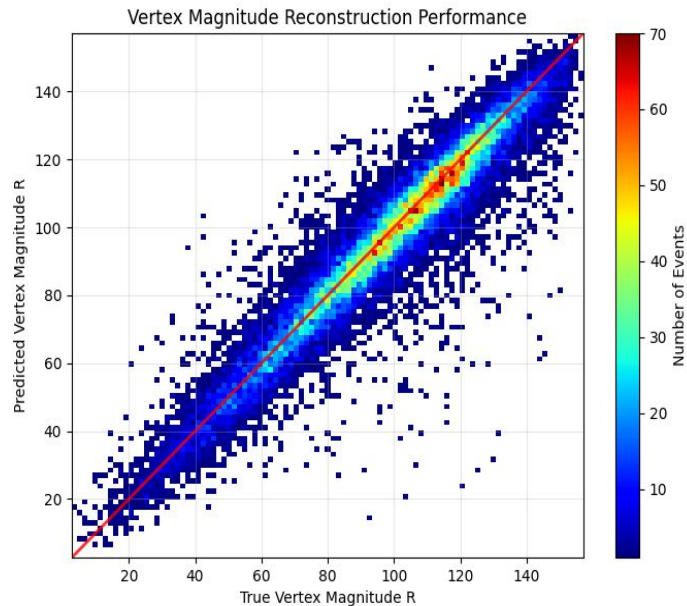
Distribution

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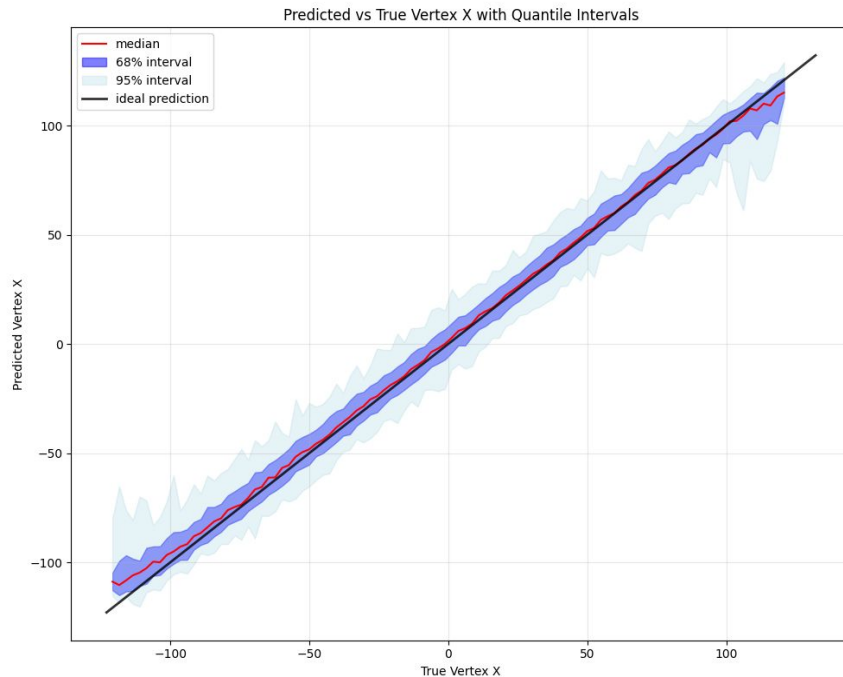
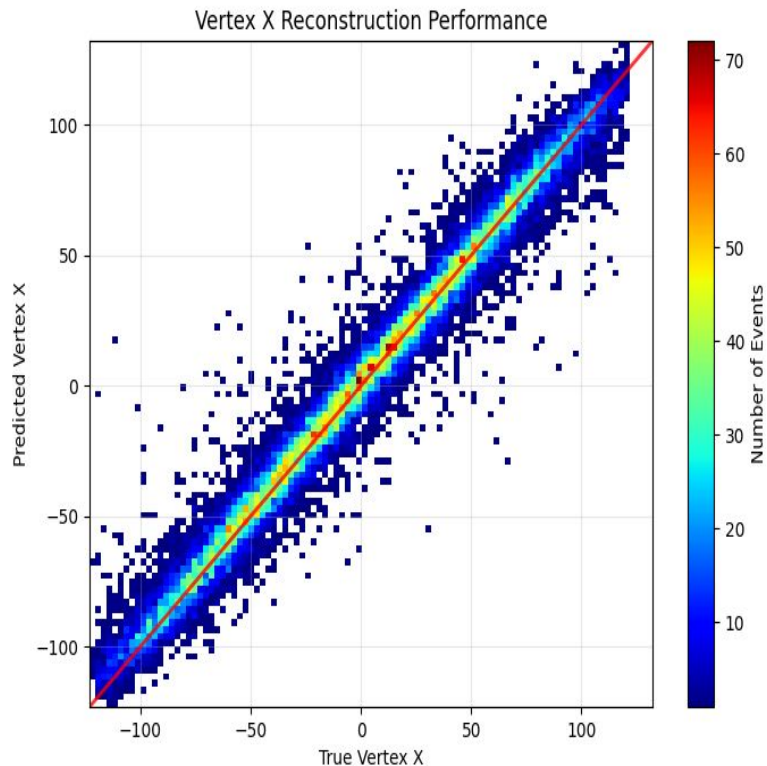
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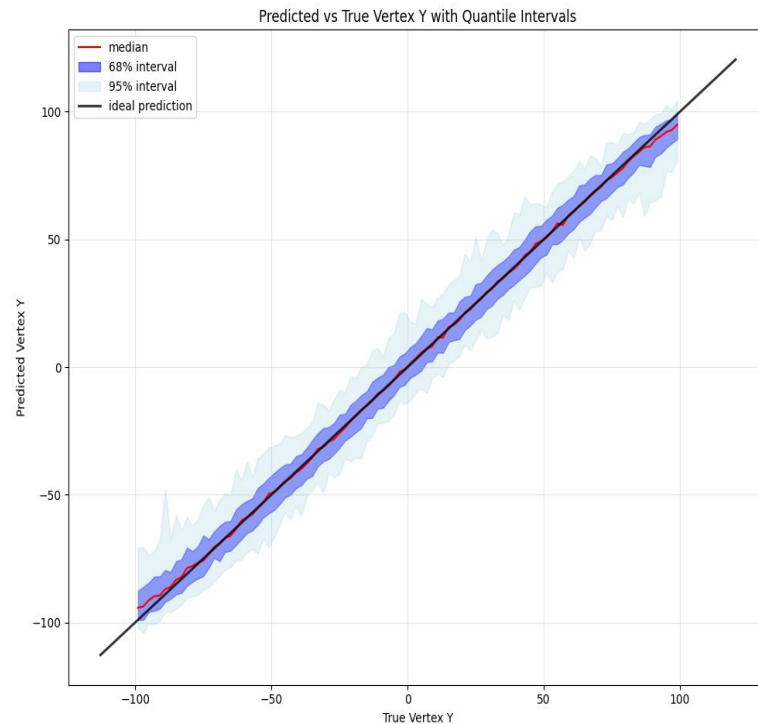
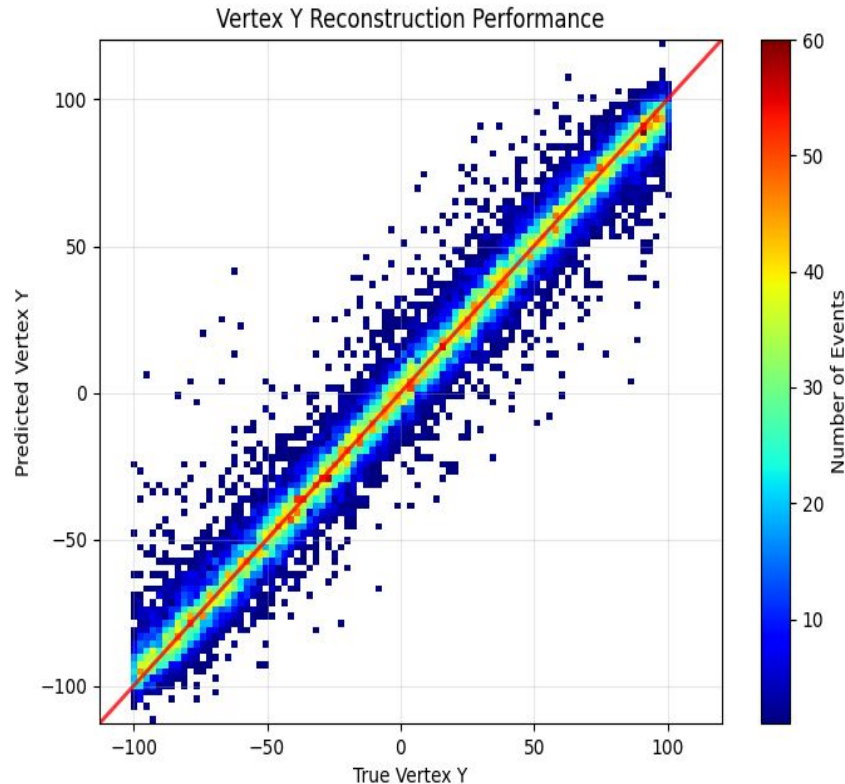
Vertex R Reconstruction and Mean Plots



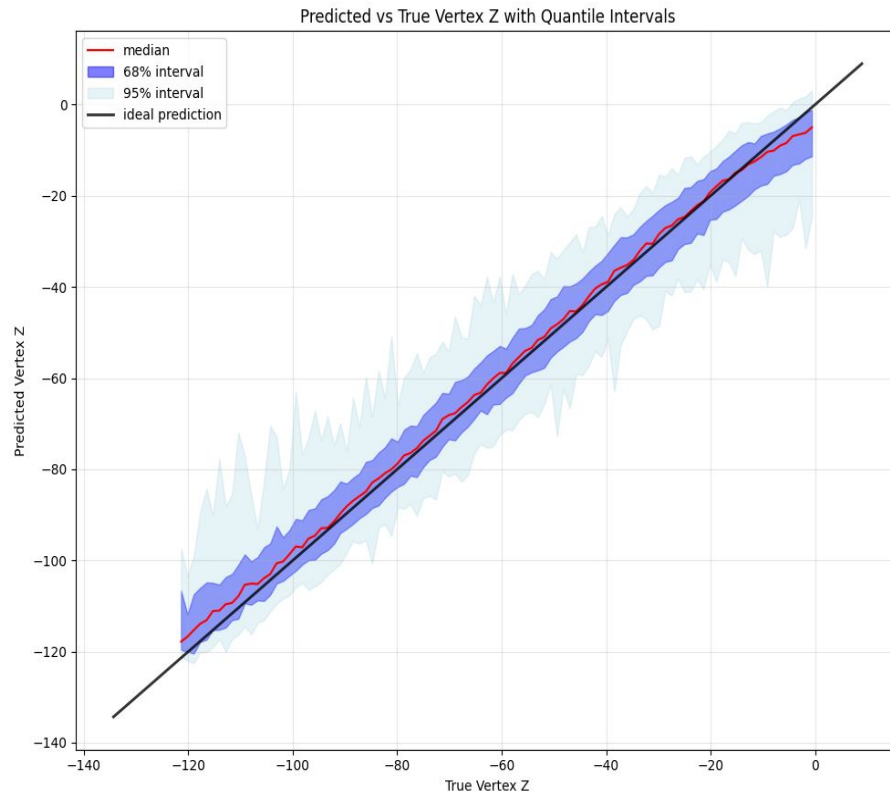
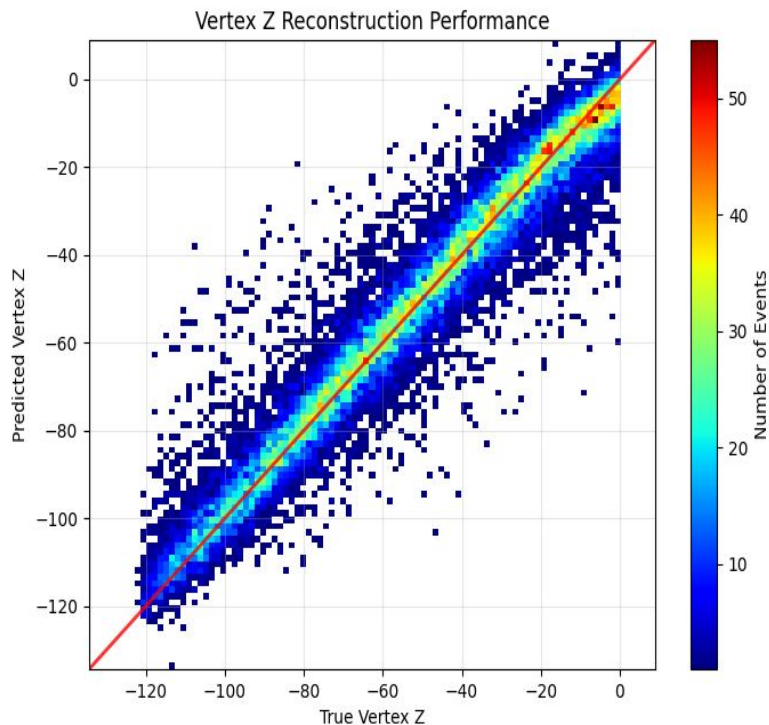
Vertex X Reconstruction and Mean plots



Vertex Y Reconstruction and Mean plots

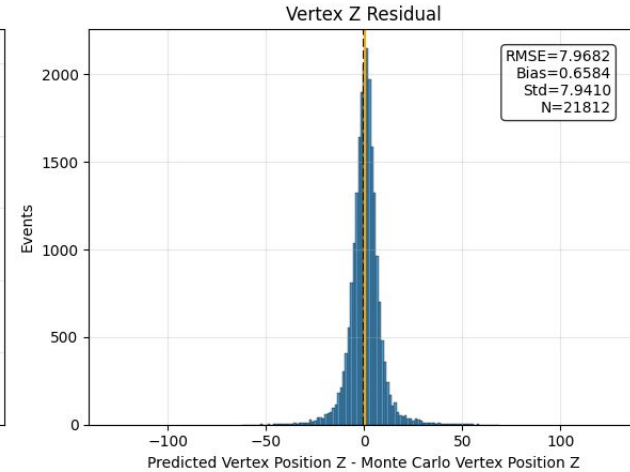
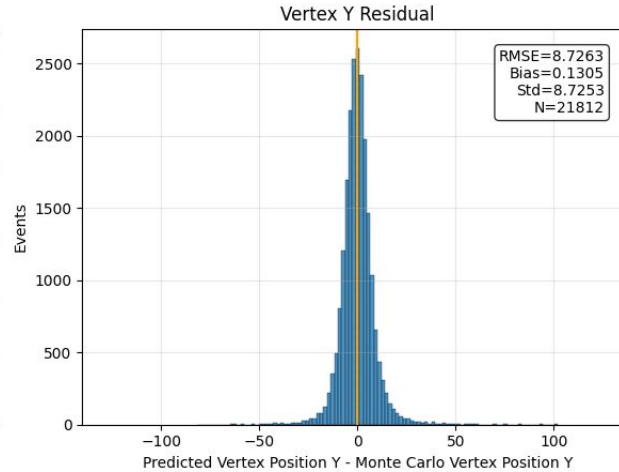
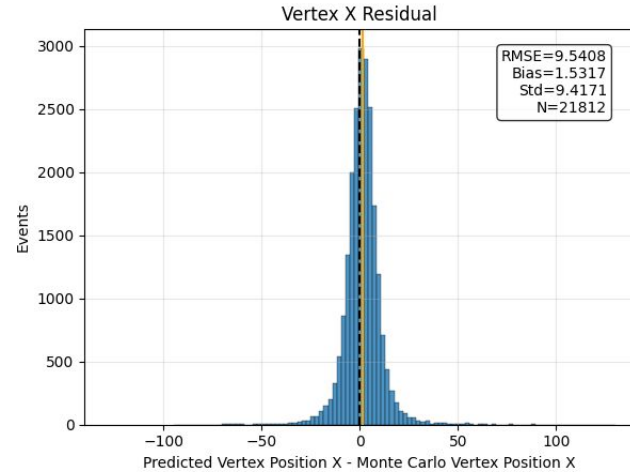


Vertex Z Reconstruction and Mean plots

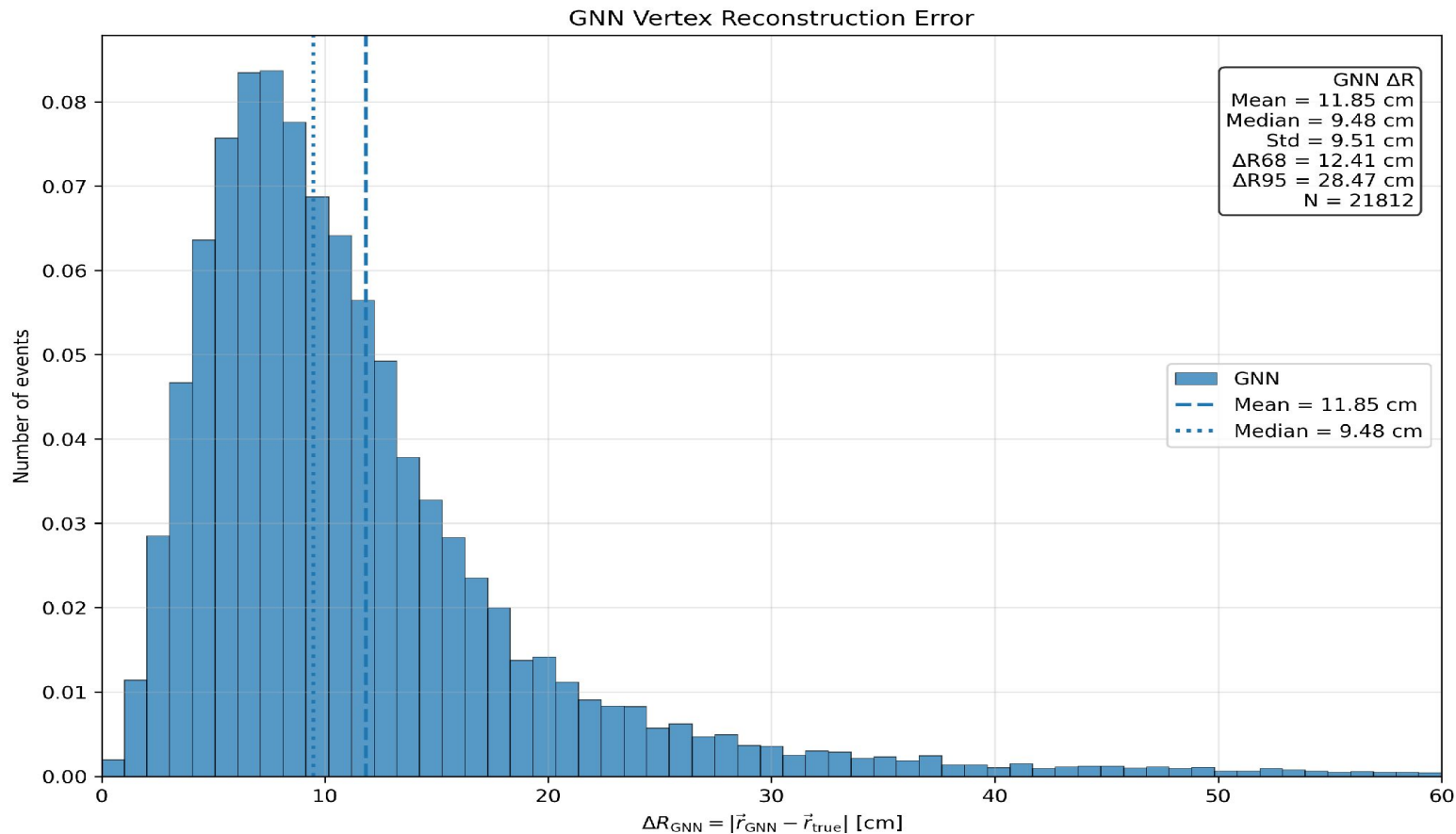


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ΔR GNN



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The containment curve shows the percentage of events with reconstruction error below a given ΔR value.

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Lower containment values indicate smaller reconstruction errors and therefore better vertex reconstruction.

